The SUD and SMI act on whether the household is employed, through the employment transitions below. There is no disability insurance in the model, so SMI brings no income of its own.[17]

Taxes. Before-tax income is taxed with the Heathcote et al. (2017) function and capital income at a flat rate, so that after-tax income is

$$y^n = \min\left\{\tilde{y},\ \tau_\ell\,\tilde{y}^{\varphi}\right\}\ +\ (1-\tau_k)\,r\,a, \tag{14}$$

where τ_ℓ scales after-tax income and so sets the level of the tax, which falls as τ_ℓ rises, and $1-\varphi$ sets its progressivity. The flat tax rate on capital income is τ_k, and r is the exogenous interest rate on savings.[18]

Transfers. Food assistance pays the maximum allotment $\underline{c}$ less the share ϱ of income. The voucher pays a holder that is eligible this period the part of its rent above a tenant contribution of κ_r of income, up to the fair market rent (FMR),

$$\begin{aligned} F &= \max\left\{0,\ \underline{c}-\varrho\,\hat{y}\right\},\\ Vch &= \mathbf{1}\{\eta=1\}\,\mathbf{1}\{\hat{y}<\kappa_V y^{\text{med}}\}\,\mathbf{1}\{h\neq h_1\}\,\min\Big\{\max\{0,\ p_r h-\kappa_r\hat{y}\},\ \max\{0,\ FMR-\kappa_r\hat{y}\}\Big\}, \end{aligned} \tag{15}$$

where $\hat{y}=\tilde{y}+ra$ is the income both programmes are means-tested on, κ_V and median income y^{med} are the eligibility rule in Section 4.9, and FMR is the average rent paid by housed households in the benchmark. A voucher holder in a shelter keeps the voucher and is paid nothing until it rents again, as long as it keeps its eligibility.

Each entering household receives a one-time endowment T_0 from bequests left by deceased households, and the government collects the rest of the estates of the dead.

4.4 The budget constraint

The household's budget constraint is as follows:

$$c+p_v v+a' \ =\ y^n+a+T_0\mathbf{1}\{j=1\}\ -\ p_r\,h\,\mathbf{1}\{h\neq h_1\}\ +\ Vch+F, \qquad c\geq F,\ v\geq 0,\ a'\geq 0. \tag{16}$$

Food assistance is paid in kind, so it cannot directly fund the drug, the rent or saving. The drug price is p_v, which is endogenous, whereas the rent per unit of housing p_r is fixed and held at its benchmark value in every simulation.[19] Treatment costs the household utility only; the government pays for it (see Section 4.11).

4.5 Housing and the homelessness loop

Housing is the four-point grid of (10): h_1 is literal homelessness, sleeping rough or in an emergency shelter, which is free, and $h_2<h_3<h_4$ are three rental sizes. Homelessness raises mortality (20), worsens both employment margins (21) and lowers the chance of recovering from the SUD (18), but does not affect SMI (Section 4.7).

4.6 Drug use, the SUD, recovery and treatment

A household that does not have the disorder, either $d=0$ (never had it) or $d=1$ (had it but recovered), acquires it ($d=2$) with probability π_d if it uses,[20]

$$\Pr(d'=2\mid d\neq 2,\,v)=\pi_d\,\mathbf{1}\{v>0\}, \tag{17}$$

[17] In the data only a minority of adults with SMI receive disability benefits on the basis of a mental disorder: about 3 million, through Social Security Disability Insurance or SSI, against 14.6 million adults with SMI in NSDUH 2023, roughly one in five.

[18] The minimum keeps the labour income tax from turning negative, since at low incomes $\tau_\ell\tilde{y}^{\varphi}$ exceeds $\tilde{y}$ itself.

[19] In earlier versions of this work, I solved the model with endogenous labour supply, rents, homeownership and house prices. These margins turned out to have a negligible effect on the results, so for conciseness I removed them and hold the rent fixed.

[20] In earlier versions of the model, I allowed the probability of developing a SUD to rise with use, but since it had little impact on the results, I removed it for simplicity.

and a household that has recovered can acquire the disorder again. A household may begin using only up to an age J_E.[21] A household with the SUD recovers, to $d = 1$, with probability

$$\pi_r(h, m, \tau) = \min\Big\{1,\ \bar{\pi}_r \left(\mathbf{1}\{h = h_1\}\,\mu_H + \mathbf{1}\{h \neq h_1\}\right)\left(\mathbf{1}\{m = 1\}\,\mu_m + \mathbf{1}\{m = 0\}\right) \times \left(\mathbf{1}\{\tau = 1\}\,g + \mathbf{1}\{\tau = 0\}\right)\Big\}, \tag{18}$$

where $\bar{\pi}_r$ is the baseline annual chance of recovering while housed, without SMI and untreated. It has three multipliers: recovery is harder in a shelter ($\mu_H < 1$), harder with SMI ($\mu_m < 1$) and more likely in treatment ($g > 1$).

Treatment τ is a choice, taken at the utility cost $\bar{\tau}^u$ in (12), and g is what it buys. Treatment does not stop the household using, and it does not remove the excess mortality of the disorder while the disorder lasts.

4.7 SMI

m is a two-state Markov chain representing SMI, mental illness severe enough to cause substantial functional impairment. It affects a small share of adults, whereas any mental illness affects roughly a quarter of adults and includes mild anxiety. Onset follows an age profile $\varsigma_m(j)$, normalised to one at age 26 and scaled by the level ω_m, and a household that carries the SUD is $\iota > 1$ times more likely to develop SMI than one that does not,

$$\Pr(m' = 1 \mid m = 0, d) = \min\left\{1,\ \omega_m\,\varsigma_m(j)\left(\mathbf{1}\{d = 2\}\,\iota + \mathbf{1}\{d \neq 2\}\right)\right\}, \qquad \Pr(m' = 0 \mid m = 1) = \lambda_m. \tag{19}$$

A household with SMI recovers from it with the probability λ_m each year, whatever its housing and whether or not it carries the SUD.

SMI then affects three objects: the chance of finding a job when unemployed, through (21); mortality, through (20), both in Section 4.8; and the SUD recovery hazard, through μ_m in (18) in Section 4.6. It does not enter the utility function.

The SMI chain does not depend on housing, because in the At Home / Chez Soi randomised trial, housing had no detectable effect on the symptoms of homeless adults with SMI (Stergiopoulos et al., 2015).

4.8 What the SUD, SMI, and homelessness cost

Death risk. Write $1 - \bar{\phi}(j)$ for the probability of dying at age j of a household that is housed and has neither the disorder nor SMI; the three multipliers below raise it for every other household. Write $q^0(j,h) = \min\{1, \left(\mathbf{1}\{h = h_1\}\,\kappa_{\phi H} + \mathbf{1}\{h \neq h_1\}\right)(1 - \bar{\phi}(j))\}$ and $q^1(j,h) = \min\{1, \kappa_{\phi d}\left(\mathbf{1}\{h = h_1\}\,\kappa_{\phi H} + \mathbf{1}\{h \neq h_1\}\right)(1 - \bar{\phi}(j))\}$ for the probability of dying without and with the disorder, respectively. In the stylized model $\kappa_{\phi d}$ was an exponent on the survival probability; here it multiplies the death hazard. The probability of dying is then

$$1 - \phi(j, h, d, m) = \min\Big\{1,\ \left(\mathbf{1}\{m = 1\}\,\kappa_{\phi m} + \mathbf{1}\{m = 0\}\right) q^0(j,h) + \mathbf{1}\{d = 2\}\left[q^1(j,h) - q^0(j,h)\right]\Big\}, \tag{20}$$

so $\kappa_{\phi H} > 1$ multiplies the death probability when a household is in a shelter, $\kappa_{\phi d} > 1$ when a household has a SUD, and $\kappa_{\phi m} > 1$ the baseline risk when a household has SMI, to which the excess risk of the disorder is added.

[21] Without this limit, using at old ages has little-to-no consequence: too little life remains for the disorder to damage, so the model produces too much old-age use. The cause is that death is certain at the last age J, so close to it a household knows how little life remains, whereas in real life the vast majority of people do not know exactly when they will die. Its value is in Appendix A. It stops only households without a SUD from starting to use; a household that already has a SUD at that age keeps using.

Unemployment risk. A job is lost with probability π^{eu} and found with probability π^{ue},

$$\begin{aligned}\pi^{eu}(y,h) &= \min\Big\{1,\ \bar{\pi}^{eu}\,\varpi(y)\,\big(1+\mathbf{1}\{h=h_1\}\,\kappa_H\big)\Big\},\\ \pi^{ue}(h,d',m') &= \min\Big\{1,\ \frac{\bar{\pi}^{ue}\,\big(1-\mathbf{1}\{h=h_1\}\,\kappa_H\big)}{\big(\mathbf{1}\{d'=2\}\,\kappa_d^{ue}+\mathbf{1}\{d'\neq 2\}\big)\big(\mathbf{1}\{m'=1\}\,\kappa_m^{ue}+\mathbf{1}\{m'=0\}\big)}\Big\},\end{aligned} \tag{21}$$

with $\bar{\pi}^{eu}$ and $\bar{\pi}^{ue}$ the baseline separation and job-finding rates, $\varpi(y)$ a factor that makes separations likelier at low productivity, and the divisors κ_d^{ue} and κ_m^{ue} making a job harder to find with the SUD and with SMI. Neither the SUD nor SMI raises separations; each only reduces the chance of regaining employment. κ_H does both for a household in a shelter.

The child's productivity penalty. A child whose parent had the disorder at any point while the child lived at home, before J_K or before the parent died if that came first, starts adult life with its productivity lowered by the proportion δ_d in (22) below, and the penalty stays with the child for life.

A child's log productivity at entry is drawn around its parent's, with persistence ρ. The innovation keeps the spread of productivity at entry the same in every generation. If the parent had the disorder at any point before the child became an adult, the penalty scales the child's productivity down for life:

$$\begin{aligned}\log y_1^{\text{child}} &= \rho\,\log y^{\text{parent}} + (1-\rho)\,\mu + \epsilon\\ &\quad + \log(1-\delta_d)\,\mathbf{1}\{\text{parent has } d\geq 1 \text{ when its child becomes an adult}\},\end{aligned} \tag{22}$$

with y_1^{child} the child's productivity in the period it becomes an adult in the workforce ($j=1$), y^{parent} the parent's productivity in the period just before its child becomes an adult, μ the mean of log productivity at entry and ϵ normally distributed with mean zero and variance $1-\rho^2$ times the variance of log productivity at entry. Because the penalised child's own productivity is lower, its own child inherits a lower draw too, so, through the persistence, the penalty ripples through the whole lineage.

4.9 The voucher and the two policies

A household is eligible for a voucher while its income is below the share κ_V of median income,

$$\hat{y} \;<\; \kappa_V\, y^{\text{med}}, \tag{23}$$

with $\hat{y}$ the income of (15). A holder keeps its voucher while it is eligible, and an eligible household without one draws one with probability π_V. Under Housing First that is the whole rule.

Under Treatment First the programme is conditional. A household fails the condition if it uses, $v>0$, or has the disorder and is not in treatment, $d=2$ and $\tau=0$. A household with a SUD must be in treatment (paid for by the government), as the executive order directs, and no household may use even while in treatment, as the guidance implementing the order requires of drug-free housing.[22]

Income-eligible holders and income-eligible households that draw a voucher are tested, at a cost c_{test} each to the government, and one that fails the condition is detected with probability p_{det}. This means that enforcement is not perfect: with probability $1-p_{\text{det}}>0$ a household that fails the condition actually gets to keep its voucher. A detected holder keeps this period's payment and loses the voucher from the next period; a detected applicant never receives it. In both cases, the household is excluded from the voucher programme for a period of time ($\eta=2$): it cannot draw a voucher whatever its income, and it returns to the lottery ($\eta=0$) with probability ψ_x each year.

None of p_{det}, ψ_x and c_{test} binds in the benchmark, where no voucher is conditional, so all three are set with the experiment in Section 6 rather than in the calibration.

[22] Executive Order 14321, Section 5(b); the guidance is *Addressing Homelessness and Addiction through "Treatment First"* (2026), p. 9, issued by the Office of National Drug Control Policy, the Department of Health and Human Services and HUD. The Housing Choice Voucher programme bars admission for current illegal drug use (Code of Federal Regulations (CFR), Title 24, Section 982.553) but does not test; Housing First here is a programme that does not test. The order also encourages the civil commitment of people with mental illness who pose a risk to themselves or others, or who live on the street and cannot care for themselves. In the model, treatment is always the household's choice: a household with a SUD that declines it loses its voucher if the screen catches it.

4.10 The household's dynamic problem

Write $\Gamma\big(s' \mid s,\, h, v, \tau, a'\big)$ for the law of motion of the state that the transitions above define, given the state and the choices. Then, for $j = 1, \ldots, J$ and with $V(J{+}1, \cdot) = 0$,

$$
\begin{aligned}
V(j,s) = \max_{\{h,\,v,\,\tau,\,a'\}} \Bigg\{ & U(X_P) + \omega_c(\xi(j)-1)\,U(X_c) + \aleph - \tau\bar{\tau}^{u} \\
& + \;\beta\,\phi(j,h,d,m)\int_{\mathcal{S}} V(j{+}1,s')\,\Gamma\big(\mathrm{d}s' \mid s,h,v,\tau,a'\big) \\
& - \;\beta\,\chi(j)\,\omega_c\,D(y,d)\Bigg\}, \qquad \chi(j) = \begin{cases} 1-\phi(j,h,d,m) & j < J_K \\ 1 & j = J_K \\ 0 & j > J_K, \end{cases}
\end{aligned}
\tag{24}
$$

subject to the budget (16), the income and transfer rules (13) to (15) in Section 4.3, and the restrictions that treatment is available only to a household with $d = 2$ and that no household begins using after J_E.

The first line is the flow of (12). The second is the household's own continuation. The third is the impact of the parent on the child's adult value, counted only when the child leaves: at J_K with certainty, and before then with the probability that the parent dies first. $D(y,d)$ is what the parent's disorder costs the child, in the child's own lifetime value,

$$
D(y,d) = \mathbb{E}\big[V(1, s_{\text{child}}) \,\big|\, y,\ 0\big] - \mathbb{E}\big[V(1, s_{\text{child}}) \,\big|\, y,\ d\big] \;\geq\; 0, \tag{25}
$$

where the child is valued as it enters at $j = 1$ with the initial states described in Section 4.13, and its productivity is drawn as in (22) in Section 4.8 from a parent with productivity y. In the first term of (25) the parent has never had the disorder, and in the second it has its actual d in the period the child leaves. If it never had a SUD, there is no charge to the parent ($D(y,0) = 0$); if it had a SUD before the child leaves ($d \geq 1$), it is charged ($D(y,d) > 0$).[23]

One weight ω_c multiplies both child terms, the flow of childhood and the value of the child's adulthood. At $\omega_c = 0$ the damage a household does to its child is entirely external; at $\omega_c > 0$ it internalises the share ω_c and only $1 - \omega_c$ remains external.

The policy functions $h(j,s)$, $v(j,s)$, $\tau(j,s)$ and $a'(j,s)$ are the dwelling, drug use, treatment and saving that solve (24) at age j and state s.

4.11 The government

Write $\Phi(j,s)$ for the distribution of the households of age j over the states s; its law of motion is in Section 4.13. The government's budget constraint is

$$
\begin{aligned}
\mathcal{R} \;=\; & \sum_{j=1}^{J} \int_{\mathcal{S}} \Big[\, \tilde{y}(j,y,e) - \min\big\{\tilde{y}(j,y,e),\, \tau_\ell\, \tilde{y}(j,y,e)^{\varphi}\big\} + \tau_k\, r\, a \\
& \qquad + \big(1 - \phi\big(j, h(j,s), d, m\big)\big)\, a'(j,s) \Big]\, \mathrm{d}\Phi(j,s) \;-\; T_0 \\
& - \sum_{j=1}^{J} \int_{\mathcal{S}} \Big[\, UI(j,y,e) + S(j,y) + F(j,s) + Vch(j,s) + c_{\text{test}}\, \mathbf{1}^{\text{test}}(j,s) + c_{\text{treat}}\, \mathbf{1}\{d=2,\, \tau(j,s)=1\} \Big]\, \mathrm{d}\Phi(j,s).
\end{aligned}
\tag{26}
$$

The first line is revenue from the labour and capital income taxes of (14). The second is the estates of the households who die, which the government collects, less the endowment T_0 it gives the unit mass of

[23] The term is a difference and not a level. Written as a level, ω_c times the child's value, the term would be worth more to a household the likelier the household is to die, because the death branch carries it. It would then pay most to the households with a SUD, for reasons that have nothing to do with the child. As a difference, the term is zero for a parent who has never had the disorder and a cost otherwise.

entrants. The third is what the government pays: UI, $UI(j,y,e) = \tilde{y}(j,y,e)\,\mathbf{1}\{j < J_R,\, e = 2\}$; social security, $S(j,y) = S(y)\,\mathbf{1}\{j \geq J_R\}$; food assistance $F(j,s)$ and the voucher $Vch(j,s)$ of (15), at the household's dwelling $h(j,s)$; under Treatment First, the cost c_{test} of each household tested, $\mathbf{1}^{\text{test}}(j,s) = 1$; and, under both policies, the cost c_{treat} of a year of treatment for each household with the SUD that takes it up, $\tau(j,s) = 1$, since treatment is publicly funded.

In the benchmark the tax parameters are set as in Section 5 and $\mathcal{R}$ is whatever surplus they produce. In every experiment the taxes are held at their benchmark values, and the probability of drawing a voucher π_V adjusts so that $\mathcal{R}$ equals its benchmark value $\bar{\mathcal{R}}$, unless the experiment fixes π_V.

4.12 The drug market

The price of the drug clears its market,

$$p_v = \nu_1 S_v^{1/\varepsilon_s}, \qquad S_v = \sum_{j=1}^{J} \int v(j,s)\,\mathrm{d}\Phi(j,s), \tag{27}$$

where ε_s is the price elasticity of supply and ν_1 is normalised so that the price is one in the benchmark, and then held fixed across experiments.[24]

4.13 The law of motion

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi(j{+}1, \mathcal{S}') \;=\; \int_{\mathcal{S}} \phi\big(j, h(j,s), d, m\big)\, \Gamma\big(\mathcal{S}' \mid s,\, h(j,s), v(j,s), \tau(j,s), a'(j,s)\big)\,\mathrm{d}\Phi(j,s), \tag{28}$$

for $j = 1, \ldots, J-1$ and every measurable set $\mathcal{S}'$ of the state space $\mathcal{S}$ of (9), with the integral over the state s of the households of age j, together with the inflow of entrants

$$\Phi(1, \mathcal{S}') \;=\; \frac{\sum_{j \leq J_K} \int_{\mathcal{S}} \chi(j)\, \mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi(j,s)}{\sum_{j \leq J_K} \int_{\mathcal{S}} \chi(j)\,\mathrm{d}\Phi(j,s)}, \tag{29}$$

where the integrals are again over s, $\mathcal{N}$ places the child according to (22) and $\chi(j)$ is the weight defined in Section 4.10. An entrant starts with the endowment T_0, is employed, has never had the disorder, holds no voucher, and carries SMI with the prevalence observed at age 26.[25]

4.14 Competitive equilibrium

A competitive equilibrium is a collection of

- value functions $\big\{\{V_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- policy functions $\big\{\{h_t(j,\cdot),\, v_t(j,\cdot),\, \tau_t(j,\cdot),\, a'_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- distributions $\big\{\{\Phi_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$; and
- prices, policies and aggregates $\big\{p_{vt},\, \pi_{Vt}\big\}_{t=0}^{\infty}$, together with $\bar{y}$, y^{med} and FMR, which carry no date because they are kept fixed at their benchmark values,

such that, at every date t,

1. the policy functions solve (24) given its terminal condition, and $\{V_t(j,\cdot)\}_{j=1}^{J}$ are the associated values;

[24] The suppliers are outside the economy: what they earn along the supply curve is counted neither in welfare nor in the budget.

[25] A disorder acquired before that age is therefore acquired in the model only after entry; however, the share of adults with a SUD still matches the data (Section 5).

2. $\Phi_{t+1}(j{+}1, \cdot)$ follows from $\Phi_t(j, \cdot)$ by the law of motion (28) together with the entrants of (29) and (22);
3. the government budget (26) holds at π_{Vt} with $\mathcal{R} = \bar{\mathcal{R}}$;
4. the drug market clears at p_{vt}, so (27) holds.

A stationary equilibrium is the special case in which all of these are constant and $\Phi_{t+1} = \Phi_t$, which is what the calibration and the long-run policy comparisons report. An experiment that holds π_V or the drug price at its benchmark value drops the budget or the drug-market condition. The sequence version is what the transition between two stationary equilibria solves.

5 Calibrating the quantitative model

The model has fifty-nine parameters, some of which vary by age. Thirty-eight are calibrated externally, from data or the literature, and twenty-one internally. The twenty-six that carry this paper's story are in Table 3, with the moment, source, data and model value behind each, and are discussed below in numbered blocks of parameters that work through similar mechanisms. The other thirty-three, common to most life-cycle models, are in Appendix A to save the reader time; its blocks continue the numbering. Three further parameters govern drug testing and exclusion from the voucher; they play no role in the benchmark and are set with the experiment in Section 6. `data/DATA_APPENDIX.md` in the replication package documents the dataset, year, universe and computation behind every estimate, and the package carries all datasets, so nothing has to be downloaded.

5.1 The twenty-six parameters that carry the story of this paper

Table 3: The twenty-six parameters that carry the story of this paper

Parameter	Description	Set	Moment and source	Data	Model	Value
Block 1: Who uses drugs						
θ	Weight on drugs in utility	Int.	Drug spending over income, Midgette et al. (2019)	16.0%	14.8%	0.165
b	Marginal utility from the first unit of drugs	Int.	Any drug use disorder, NSDUH 2024	8.90%	8.96%	0.0054
π_d	Chance that a year of use leads to a SUD	Int.	Share of users with a SUD, NSDUH 2024	41.3%	42.6%	0.0345
$\kappa_{\phi d}$	Increased mortality with a SUD	Int.	Income of drug users over average, NSDUH 2024	74%	62%	4.00
$\aleph$	Flow value of being alive	Int.	VSL, Viscusi and Aldy (2003)	80	81	87.6
Block 2: Costs of homelessness and vouchers						
κ_d^{ue}	How much less likely it is to regain employment with a SUD	Int.	SUD share of the homeless, Gutwinski et al. (2021)	21.7%	20.2%	3.78
h_1	Service flow of homelessness	Int.	Homelessness band, HUD PiT 2023 and Burt et al. (2001)	0.40–1.00%	0.62%	0.0216
$\kappa_{\phi H}$	Mortality when homeless	Ext.	Meyer et al. (2023)	1.60	n.a.	1.60
κ_H	Employment penalty of homelessness	Ext.	Desmond and Gershenson (2016)	0.20	n.a.	0.20
$\underline{c}$	Food stamp maximum allotment	Int.	SNAP outlays over output	0.42%	0.43%	0.0879
π_V	Chance of drawing a voucher	Int.	Recipients over eligible, HUD 2023	26.6%	27.5%	0.0675
T_0	Endowment at entry to adulthood ($j = 1$)	Int.	Share of homeless aged 25 to 34, HUD PiT 2024	26.0%	26.4%	0.111
Block 3: SMI chain						
ι	SUD multiplier on SMI onset	Ext.	Danish registers, Nielsen et al. (2017)	2.00	n.a.	2.00
λ_m	Annual chance of recovering from SMI	Ext.	SMI ten-year recovery rate, Åsbø et al. (2022)	50%	n.a.	0.066
ω_m	Level of the SMI onset profile	Int.	SMI share of the population, NSDUH 2024	4.82%	4.83%	0.0037
κ_m^{ue}	How much less likely it is to regain employment with SMI	Int.	SMI share of the homeless, Gutwinski et al. (2021)	25%	24.7%	7.71
μ_m	SUD recovery multiplier for SMI	Ext.	Dual-diagnosis relapse, Bradizza et al. (2006)	0.82	n.a.	0.82
$\kappa_{\phi m}$	Mortality multiplier for SMI	Ext.	Olfson et al. (2015)	2.50	n.a.	2.50
Block 4: Recovery and treatment						
$\bar{\pi}_r$	Baseline SUD recovery rate	Ext.	SUD ten-year remission rate, Vaillant (1995)	22%	n.a.	0.025
μ_H	SUD recovery when homeless	Int.	Abstinence, housed/homeless, Milby et al. (2005)	1.47	1.45	0.696
g	Treatment effect on recovery	Ext.	Remission on buprenorphine, Paul et al. (2024)	2.00	n.a.	2.00
$\bar{\tau}^u$	Utility cost of treatment	Int.	Treatment take-up, NSDUH 2024	24.5%	24.9%	4.49
c_{treat}	Public cost of a year of treatment	Ext.	Residential treatment episode, Roebuck et al. (2003)	\$9,426	n.a.	0.133
Block 5: Externalities						
ω_c	Parent's weight on its child	Int.	Two-adult over lone-parent households' use, less one, NSDUH 2024	17.5%	17.7%	0.085
δ_d	Earnings the child loses because of a parent's SUD	Ext.	Earnings lost in utero, Nilsson (2017)	0.24	n.a.	0.24
Block 6: Drug market						
ε_s	Price elasticity of drug supply	Ext.	Mace et al. (2020)	1.46	n.a.	1.46

Notes: the Model column is the Housing First benchmark. n.a. marks a parameter set from outside the model, which has no moment of its own to reproduce.

Block 1: Who uses drugs. The five parameters in this block set participation and, more importantly, *who* participates. Neither policy acts on them directly; they decide how much there is for a policy to act on.

θ, the weight on drugs in utility. It is the drug's weight in the bundle, so it mostly sets how much a household that uses spends on drugs, but a higher weight also makes using at all worth more. As it

rises, use therefore rises on both the extensive and the intensive margin, and with it the externalities and what either policy has to act on. It is calibrated internally, at 0.17, to aggregate drug spending by users over their aggregate income, 16.0 percent, from retail drug spending in Midgette et al. (2019) and, in NSDUH 2024, the incomes of adults aged 26 and over who use drugs.

b**, the marginal utility from the first unit of drugs.** The marginal value of the first unit of drugs goes as θ/b, so a larger b makes the first units worth less and a household with little to spend buys none. As b falls, the marginal utility from the first unit rises, and so do overall drug use and the externalities that policy can affect. It targets a participation rate of 8.90 percent, the NSDUH 2024 rate of a drug use disorder of any severity among adults aged 26 and over.[26]

π_d**, the chance that a year of use leads to a SUD.** This is what makes using *risky*: the step from drug use to a disorder. As it rises, the externalities rise, and the moral hazard of Housing First turns an extra voucher into more SUD, which Treatment First then aims to prevent. It targets the share of users who have a SUD, 41.3 percent among adults aged 26 and over in NSDUH 2024.

$\kappa_{\phi d}$**, increased mortality with a SUD.** $\kappa_{\phi d}$ sets how much a SUD raises the chance of dying, and mortality truncates a user's value of life. Because this is an uninsurable risk, a rise in $\kappa_{\phi d}$ hits the households with the highest values of life hardest, which reverses the income gradient; as the stylized model of Section 3 shows, this makes it a key parameter. It is calibrated internally, at 4.00, to the income of users relative to the average, a value within the range of the mortality rate ratios of a SUD in the literature: 3.91 in the Danish registers (Plana-Ripoll et al., 2019) and 6.6 among Norwegians in specialist care for a SUD (Heiberg et al., 2018). Users earn 62 percent of the average in the model against 74 percent in the data.

$\aleph$**, the flow value of being alive.** $\aleph$ is the other half of the mortality mechanism: $\kappa_{\phi d}$ sets how much a SUD raises the chance of dying, and $\aleph$ what the life lost is worth. As it rises, dying costs more and use falls. It targets a VSL of 80 times average labour income (Viscusi and Aldy, 2003).[27]

Block 2: Costs of homelessness and vouchers. Both policies work on this block, from opposite ends: each parameter says how welfare reducing a state is, and so what a voucher and its removal are worth in it.

κ_d^{ue}**, how much less likely it is to regain employment with a SUD.** An unemployed household with a SUD finds work at the rate $\bar{\pi}^{ue}$ divided by $\kappa_d^{ue} > 1$. As κ_d^{ue} rises, the expected unemployment spells of households with a SUD lengthen and they are likelier to lose their housing: this is the link from the disorder to the street, and it shapes how much Treatment First deters. It is calibrated internally to the SUD share of the homeless, 21.7 percent in Gutwinski et al. (2021).

h_1**, the service flow of homelessness.** This parameter sets how welfare reducing the outside option is, which affects the value of life: a household with little to lose from dying faces a smaller mortality cost of using. As h_1 rises, a shelter is less welfare reducing, a voucher is worth less and Treatment First has less to act on. It targets a homelessness rate within 0.40 to 1.00 percent of households: the floor is HUD's 2023 PiT count, which counts a single night; the ceiling is the share homeless at some point over a year, from Burt et al. (2001). Its value is 53 percent of the smallest rental (Appendix A), and the model delivers an aggregate homelessness rate of 0.62 percent.

$\kappa_{\phi H}$**, mortality when homeless.** This parameter sets how much more likely a homeless household is to die. As it rises, the cost of homelessness rises, a voucher is worth more and Treatment First's deterrence is stronger. It is set externally to multiply baseline mortality by 1.60 at every age, the mortality ratio for homeless adults against poor housed adults in Meyer et al. (2023).

κ_H**, the employment penalty of homelessness.** This parameter sets how much likelier a household in a shelter is to lose its job and how much less likely it is to find one: separations are $1 + \kappa_H$ times

[26] A user in the model is therefore a household whose use of any illicit drug, marijuana among them, amounts to a disorder, mild included. A household with a SUD is one whose disorder is moderate or severe (next item). Use that never reaches a disorder is not modelled.

[27] No household has a negative value of life in equilibrium, so death is never optimal. Because the value of being alive enters utility additively and utility is concave, the VSL rises faster than consumption, roughly with its square at the model's risk aversion, as in Hall and Jones (2007). Estimates of the income elasticity of the VSL are lower across people at a point in time, 0.5 to 0.6 in Viscusi and Aldy (2003), than over time, 1.5 to 1.7 in Costa and Kahn (2004).

as frequent and job finding $1 - \kappa_H$ times as fast. Housing First recovers that employment directly, whereas Treatment First does so only as the fall in SUD lowers unemployment and homelessness over time. As κ_H rises, homelessness costs more, and both policies have more to act on. It is set externally at 0.20, from Desmond and Gershenson (2016).

$\underline{c}$**, the food stamp maximum allotment.** This parameter sets how low consumption can go for the poorest households, with no income, assets or voucher, and so where utility is bounded from below. As it rises, the value of being alive rises, mortality costs more and Treatment First has more to deter. It is calibrated internally to food stamp spending over output, 0.42 percent: outlays from the United States Department of Agriculture (USDA) over GDP.

π_V**, the chance of drawing a voucher.** This is the policy dial itself, and it holds the whole trade-off. Raising it houses more households, which reduces the *childhood homelessness externality*, and makes life worth more, which makes Treatment First's deterrence more effective. In the benchmark it is calibrated internally, at 0.068, so that the share of income-eligible households who hold a voucher is aligned with the 26.6 percent in HUD's Picture of Subsidized Households, 2023.

T_0**, the endowment at entry to adulthood** ($j = 1$)**.** A child who becomes an adult starts adult life with T_0 times average earnings, which sets how likely a young adult is to end up in a shelter in its first years. As it rises, fewer young adults become homeless, and Treatment First affects behaviour less in the child-rearing years, when parents' choices affect their children. Its moment is the share of the homeless aged 25 to 34, 26.0 percent in HUD's 2024 PiT count.

Block 3: SMI chain. This block sets how much further Treatment First reaches beyond the disorder it prevents: a SUD raises the chance of SMI, and SMI lowers employment and raises mortality.

ι**, the SUD multiplier on SMI onset.** The hinge of the block: without it, SMI is exogenous to everything a policy does. As it rises, a SUD causes more SMI, and Treatment First's return rises, because preventing the disorder now prevents some SMI too. It is set externally at 2.00, between the hazard ratio of 1.34 for any SUD that Hjorthøj et al. (2018) estimate for the conversion from schizotypal disorder to schizophrenia and the 6.04 for any substance abuse that Nielsen et al. (2017) estimate for the onset of schizophrenia, both in the Danish registers.

λ_m**, the annual chance of recovering from SMI.** This parameter sets how long an episode of SMI is expected to last. As it rises, SMI is less persistent, and Treatment First's deterrence matters less, since the SMI spells a SUD can set off are shorter on average. It is set externally at 0.066, the annual hazard that matches the ten-year clinical recovery rate of 50 percent for bipolar disorder in the first-episode psychosis cohort of Åsbø et al. (2022).

ω_m**, the level of the SMI onset profile.** This parameter scales the chance of developing SMI at every age, and so how many households have it. As it rises, a SUD's effect on SMI is larger, and Treatment First has more to act on. It is calibrated internally to the share of the population with SMI, 4.82 percent, the prevalence in NSDUH 2024.

κ_m^{ue}**, how much less likely it is to regain employment with SMI.** The same as κ_d^{ue} for SMI: a household with SMI finds work at the rate $\bar{\pi}^{ue}$ divided by $\kappa_m^{ue} > 1$. As it rises, expected unemployment spells with SMI lengthen and SMI costs more. Its moment is the SMI share of the homeless, 25 percent in Gutwinski et al. (2021), since a household out of work longer is likelier to become homeless.

μ_m**, the SUD recovery multiplier for SMI.** The chain's second direction: a SUD raises the chance of SMI, and SMI makes a SUD more persistent. As it falls, recovering from a SUD is less likely, and Treatment First has more to overcome. It is set externally at 0.82 from the dual-diagnosis relapse rates in Bradizza et al. (2006).

$\kappa_{\phi m}$**, the mortality multiplier for SMI.** The chain's last link, through which SMI shortens life. As it rises, more households with SMI die, and Treatment First has more to work on, because less drug use means less SUD, less SMI and fewer deaths. It is set externally at 2.50, the middle of the standardised mortality ratios for SMI in Olfson et al. (2015).

Block 4: Recovery and treatment. Both policies work through this block: Treatment First encourages seeking treatment, and Housing First buys recovery by housing people, since recovery is likelier while housed. The first two parameters set how much recovery there is to buy, the next two how much of it treatment reaches, and the last what treatment costs the government.

$\bar{\pi}_r$**, the baseline SUD recovery rate.** This parameter is the annual chance that a housed household recovers from a SUD, the level everything else in the block scales. As it rises, disorders are expected to end more quickly on their own, and there is less for either policy to do. It is set externally at 0.025, the annual hazard implied by the ten-year remission rate in Vaillant (1995).

μ_H**, SUD recovery when homeless.** This parameter sets how much less likely recovery is in a shelter, and so how much recovery a voucher buys beyond shelter. As it falls, housing buys more recovery and Housing First is more effective in curing a SUD; near one, Housing First has little effect. It targets the effect of housing on recovery in Milby et al. (2005), who find that housing improves recovery chances by 47.3 percent.

g**, the treatment effect on recovery.** The instrument through which Treatment First reaches households that already have a SUD, which matters most in the short run. As it rises, treatment is likelier to end a SUD, and Treatment First has a larger margin to act on. It is set externally at 2.00, which is how much medication for opioid use disorder raises remission: rates near double those without it (Paul et al., 2024), and randomised trials of agonist medication against placebo find at least as large an effect on abstinence (Fudala et al., 2003).

$\bar{\tau}^u$**, the utility cost of treatment.** Take-up is a choice, and this utility cost is what makes households decline a treatment that would raise their chance of recovering. As it rises, fewer take treatment up, and Treatment First induces fewer households with a SUD to seek it. It is calibrated internally to treatment take-up: 24.5 percent of adults with a SUD received substance use treatment in the past year in NSDUH 2024.

$c_{\mathbf{treat}}$**, the public cost of a year of treatment.** This parameter is what the government pays for each household with a SUD that takes up treatment, under both policies. It ties treatment to the government's budget. Treatment First raises take-up, since entering treatment and stopping is how a household with a SUD keeps its voucher, so part of what the screen saves goes on treatment rather than on vouchers for others. As c_{treat} rises, Treatment First frees less for vouchers and π_V rises less. It is set externally to \$9,426 a year, 8.4 percent of average earnings, the cost of a residential treatment episode in Roebuck et al. (2003).

Block 5: Externalities. These two parameters decide what is an externality and how large it is.

ω_c**, the parent's weight on its child.** A parent bears the share ω_c of what its choices cost its child, in childhood and through the productivity penalty. As it rises, the parent counts more of the child's loss, so the *budget, earnings and childhood homelessness externalities* left for either policy to correct are smaller. It is identified by a counterfactual moment: drug use when the child is given no weight, against drug use at ω_c, with every other parameter unchanged.[28] The data moment holds the household fixed as well. In NSDUH 2024, an unmarried adult living with one other adult and no child was 17.5 percent more likely to have used illicit drugs in the past year than a lone parent living with one child. The lone parents are reweighted to the other group's age, sex, marital status (never married, divorced or separated, or widowed) and family income; with two members in each household, equal family income is equal income per person.[29]

[28] With $\omega_c = 0.69$, the weight on children that Nishiyama (2000) estimates and the upper end of the literature, and with b re-matched to drug use, Treatment First still raises a newborn's welfare, by 3.18 percent of lifetime consumption, against 2.57 in the benchmark. Less of that gain is an externality, however: 23 percent of it, against 76 percent in the benchmark (the last column of Table 6), because a parent with this weight counts most of what its choices cost its child, 1.64 percent of lifetime consumption against 0.19. Quantitative models in the dynastic tradition of Barro and Becker (1989) typically use lower weights (Lee and Seshadri, 2019; Abbott et al., 2019).

[29] This moment counts any past-year use of an illicit drug, a broader measure than the model's, because too few lone parents in the sample have a disorder for the comparison to give a statistically reliable measure. It is therefore used only for how much use responds, in proportion, and not for its level: the model matches it as the proportional rise in its own users.

δ_d, **earnings the child loses because of a parent's SUD.** The child inherits not the disorder but a worse productivity draw, which makes it likelier to be poor, to use, to acquire a SUD and SMI, to be homeless and to die earlier. As it rises, with $\omega_c < 1$, the *earnings externality* grows, which favours whichever policy prevents more drug use: Treatment First by deterring the parent, Housing First by making recovery likelier once housed. It is set externally at 0.24, the earnings loss, at age 32, that Nilsson (2017) measures for children of mothers under 21 exposed in utero to a Swedish alcohol experiment. Section 2 explains why the value is conservative. No study measures the earnings cost of a parent's disorder after birth directly; Nilsson's is the closest causal estimate, and the SUD odds ratio of Block 6 of the validation (Section 5.2) is the indirect check on its size.[30]

Block 6: Drug market. One parameter makes the drug price endogenous, so that it moves with demand.

ε_s, **the price elasticity of drug supply.** This parameter sets the size of a general-equilibrium offset that runs against Treatment First: deterring users lowers demand and the price, and a cheaper drug draws other households in. As it rises, supply is more elastic and the price moves less, so the offset shrinks. I set it to 1.46, the estimate of Mace et al. (2020) for cannabis on Washington State administrative data.[31]

5.2 Validation

Table 4 compares the model with moments it was not fitted to, in numbered blocks, and each block says why it matters for the experiments.

Block 1: Homelessness. The first row gives each age band's share of the homeless; the youngest band is the target of T_0, the other four are free. The model puts 37.9 percent of the homeless at ages 35 to 44, against 27.2 in HUD's PiT count, and 11.5 percent at 55 and over, against 25.9. Of the homeless in the model, 21.4 percent are employed, against 46 in the data, and their income is 6.5 percent of the average, against 19.1 in Meyer et al. (2024). As in the data, most of the homeless are of child-rearing age and poor, so the model places Treatment First's margin, the voucher a household holds or loses, at the ages where it reaches children.

Block 2: Who uses and who has a SUD. Users in the model are somewhat younger than in the data: 67 percent of them are 26 to 34, against 45 in NSDUH 2023, and their mean age is 34.6, against 39.6. Unemployment among households with a SUD is 14.5 percent, against 16.2 in the data. Households with SMI are 2.64 times more likely than others to use drugs, against 2.44, and excess deaths due to SUD account for 14.5 percent of deaths at ages 25 to 44, against 20.0 percent from drug poisonings in the Centers for Disease Control and Prevention (CDC) WONDER database. Both policies reach a child through its parent's decisions, and users are mostly of child-rearing age, in the model as in the data. And since households with SMI use drugs and are homeless at higher rates than the rest, as in the data, the model also captures that Treatment First's screen falls disproportionately on them: they are the likeliest to fail it.

Block 3: Housing and vouchers. In the model, 58.7 percent of housed households spend over thirty percent of their income on rent, against 43.2 percent of renters in the SCF. A voucher holder pays 39.0

[30] The earnings externality is only 18 percent of the welfare gain from Treatment First (Section 6.3), so the ranking of the two policies does not rest on the penalty's size.

[31] To the best of my knowledge, there is no direct estimate for an illicit drug, and cannabis is the closest market there is: it remains a Schedule I substance federally, so even a state-licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces. Mace et al.'s estimate is a short-run one, however: it is the response of Washington's legal retailers in the month after Oregon opened its own retail market in October 2015 (2.03 and 1.76 with windows of two and three months). It also comes from a market whose producers' licences capped their growing area. The long-run elasticity of an illicit supply, which faces no such cap, is likely higher, and studies of illicit drugs take it to be infinite (Becker et al., 2006).